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DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING (EEE)
ELECTRONIC DEVICES LAB

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Experiment No: 02
Name of the Experiment:
Study of Diode Rectifiers.

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Abstract:

This lab is about understanding how diodes work in rectifier circuits to change AC (alternating current) into DC (direct current). We will focus on two types of rectifiers half-wave and full-wave and see how they do this conversion. Rectifiers are important because they make AC power usable in DC form for many electronic devices.

We will start by setting up half-wave and full-wave rectifier circuits and observing how they work. By measuring things like output voltage and ripple, we will learn about the efficiency and stability of each type. We'll also use a simulator to predict how the circuits will behave before building them. By comparing the simulated results with real-world measurements, we can see how well theory matches practice. This process will help us better understand rectifiers, their strengths and weaknesses, and their role in electronic systems.

Theory

A diode rectifies an AC voltage so that it can be smoothed and converted into a DC voltage. A rectifier, however, can produce a constant or variable DC voltage. A diode rectifier can produce a fixed DC voltage whereas an SCR can produce a variable DC voltage. Diode rectifiers are of the following types:

1. Half-wave rectifier.
2. Full-wave bridge rectifier.
3. Center-tapped Full-wave rectifier.

A rectifier, however, cannot produce a smooth DC voltage. So, the rectification block that makes the output DC voltage a smooth one follows a filter circuit. In this case, the capacitor acts as a smoothing filter so that the output is nearly a DC voltage. The filtering is not perfect; there will be a remaining voltage fluctuation known as a ripple, on the output voltage.

The half-wave voltage signal is normally established by a network with a single diode that has an average or equivalent DC voltage level equal to 31.8% of the peak voltage, whereas the full-wave rectified signal has twice the average or DC level of the half-wave signal, or 63.6% of the peak value.

Working Principle of Half-Wave Rectifier

In a half-wave rectifier, only half a cycle of applied AC voltage is used. Another half cycle of AC voltage (negative cycle) is not used. Only one diode is used which conducts during the positive cycle. The circuit diagram of a half-wave rectifier without a capacitor is shown in Fig. 1.

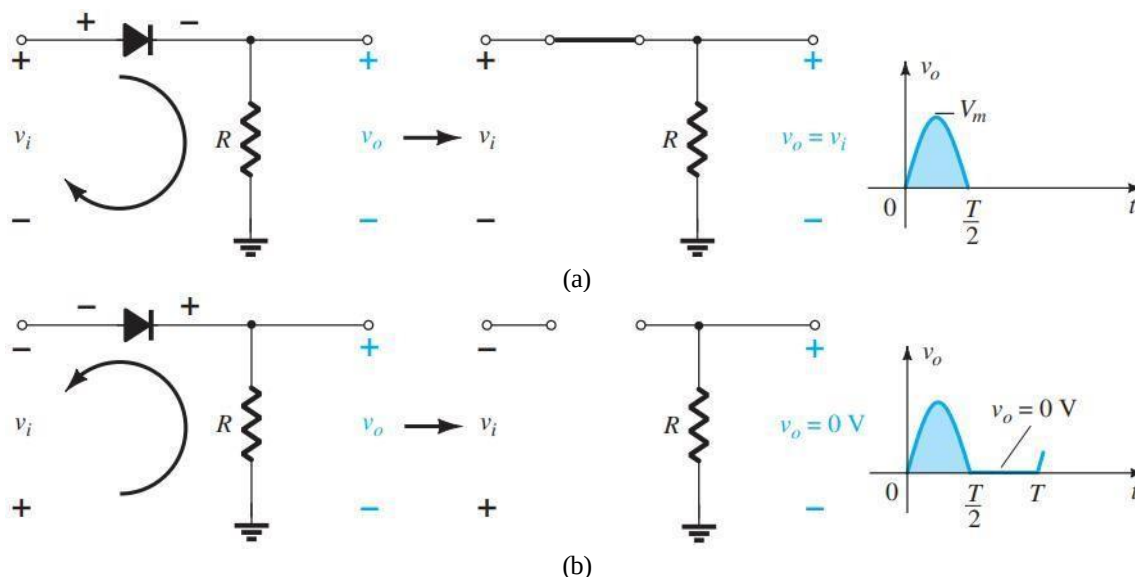


Figure 1: Half-Wave Rectification: (a) conduction in the positive half-cycle from 0 to $T/2$; (b) no conduction in the negative half-cycle from $T/2$ to T .

During the positive half cycle of the input voltage anode of the diode is positive compared with the cathode. The diode is in forward bias and current passes through the diode and a positive cycle develops across the load resistance R as shown in Fig. 1 (a). During the negative half cycle of input voltage, the anode is negative with respect to the cathode and the diode is in reverse bias. No current passes through the diode and hence the output voltage is zero across the load resistance R as shown in Fig. 1 (b). The complete wave shape for two cycles is shown in Fig. 2. The average value of the half-wave rectified voltage signal over a full period (T) is given by

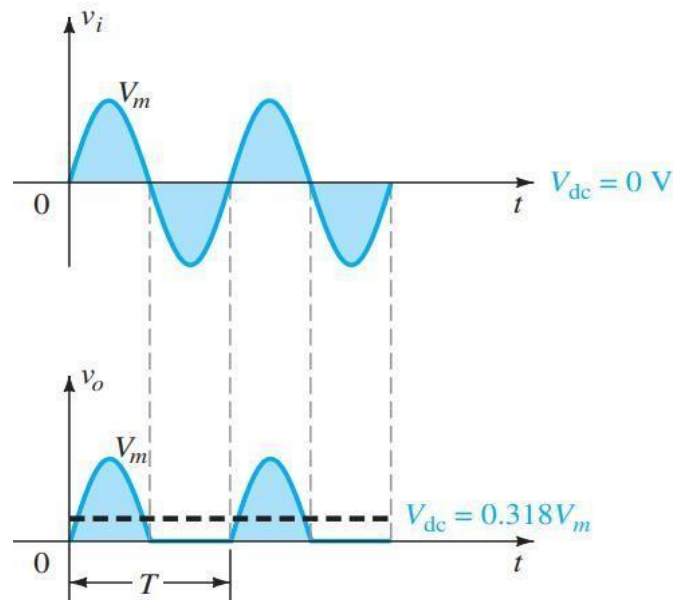


Figure 2: Half-wave rectified signal for two cycles.

Working Principle of Full-Wave Rectifier:

The DC level obtained from a sinusoidal input can be improved by 100% using a process called full-wave rectification. The bridge rectifier is a circuit that converts an AC voltage to a DC voltage using both half cycles of the input AC voltage. The most familiar network for performing such a function is a bridge rectifier as shown in Fig. 3 with its four diodes in a bridge configuration. The AC input voltage is applied to the diagonally opposite ends of the bridge. The load resistance is connected between the other two ends of the bridge.

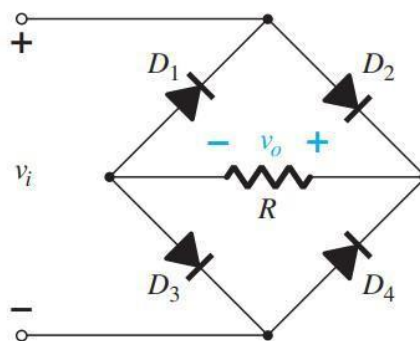


Figure 3: Full-wave bridge rectifier.

During the period $t = 0$ to $T/2$, that is, in the positive half-cycle, the polarity of the input is shown in Fig. 4. The resulting polarities across the ideal diodes are also shown in Fig. 4 to reveal that D_2 and D_3 are conducting, whereas D_1 and D_4 are in the “off” state. The net result is the configuration of Fig. 4, with its indicated current and polarity across the load resistance, R . Since the diodes are ideal, the load voltage is $v_o = v_i$, as shown in the same figure. The bridge rectifier circuit is shown in Fig. 3.

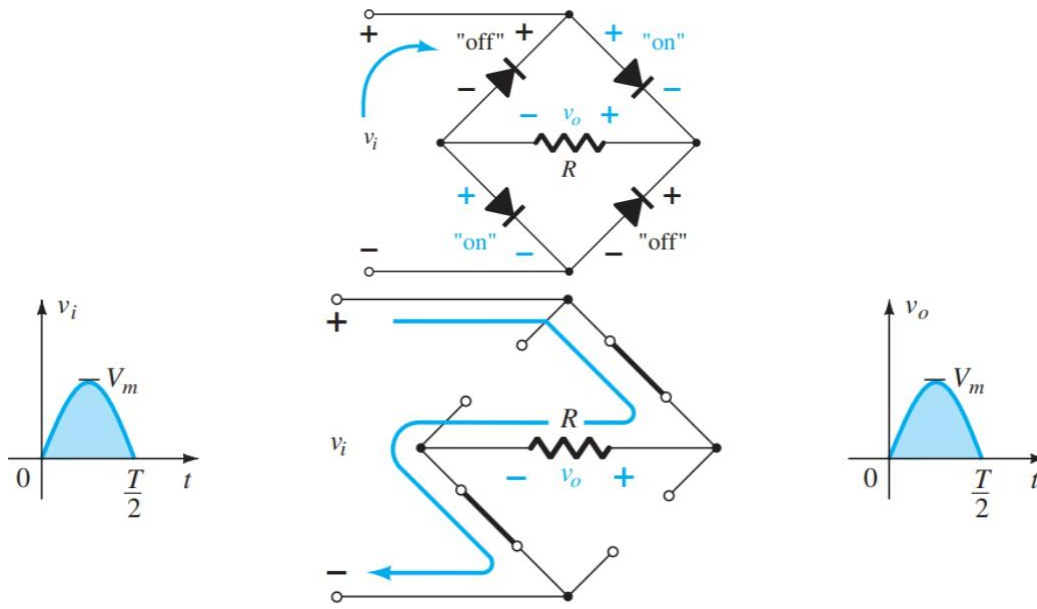


Figure 4: During the positive half-cycle of the input signal, D2 and D3 diodes are forward-biased and conduct current, while D1 and D4 diodes are reverse-biased and hence conduct no current.

For the negative half-cycle of the input, the conducting diodes are D1 and D4, resulting in the configuration of Fig. 5. The important result is that the polarity across the load resistor, R , is the same as in Fig. 5, establishing a second positive pulse, as shown in Fig. 5.

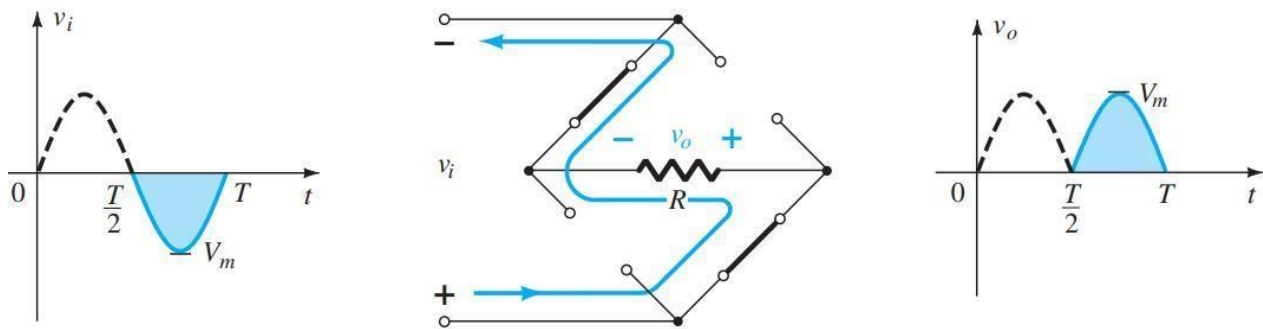


Figure 5: During the negative half-cycle of the input signal, D1 and D4 diodes are forward-biased and conduct current, while D2 and D3 diodes are reverse-biased and hence conduct no current.

Over one full cycle, the input and output voltages will appear as shown in Fig. 6. The average value of the full-wave rectified voltage signal over a full period (T) is given by

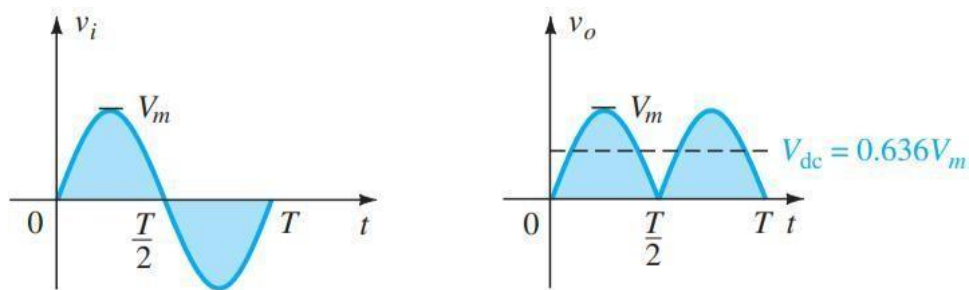


Figure 5: Input and output waveforms for a full-wave rectifier.

Working Principle of Center-Trapped Full-Wave Rectifier:

A second popular full-wave rectifier is a center-tapped rectifier. It is a type of full-wave rectifier that uses two diodes connected to the secondary of a center-tapped transformer, as shown in Fig. 6. The input voltage is coupled through the transformer to the center-tapped secondary. Half of the total secondary voltage appears between the center tap and each end of the secondary winding as shown in Fig. 6 with only two diodes but requiring a center-tapped (CT) transformer to establish the input signal across each section of the secondary of the transformer.

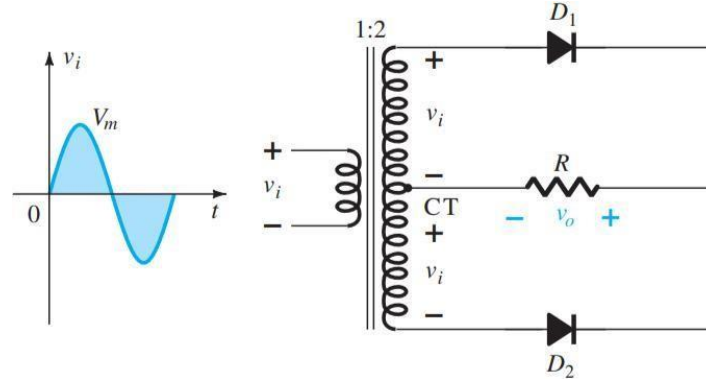


Figure 6: Center-tapped transformer full-wave rectifier.

During the positive half-cycle of v_i applied to the primary of the transformer, the network will appear as shown in Fig. 7 with a positive pulse across each section of the secondary coil. D1 assumes the short-circuit equivalent and D2 the open-circuit equivalent, as determined by the secondary voltages and the resulting current directions. The output voltage appears as shown in Fig. 7.

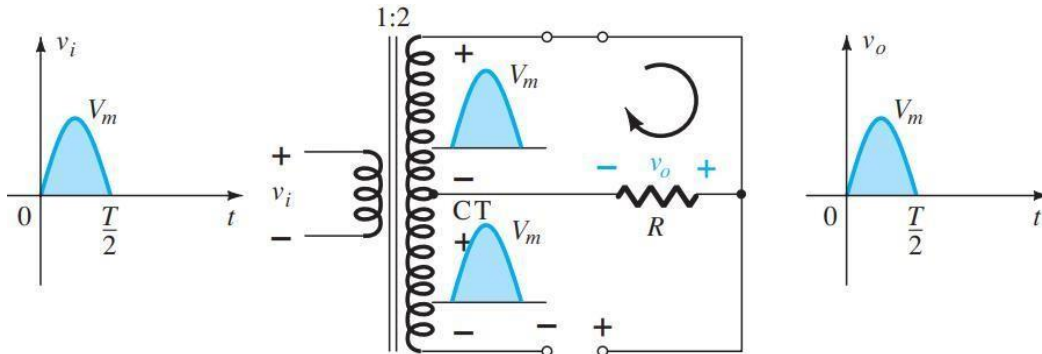


Figure 7: Network conditions for the positive region of v_i .

During the negative half-cycle of v_i applied to the primary of the transformer, the network will appear as shown in Fig. 8 with a positive pulse across each section of the secondary coil. D2 assumes the short-circuit equivalent and D1 the open-circuit equivalent, as determined by the secondary voltages and the resulting current directions. The output voltage appears as shown in Fig. 8.

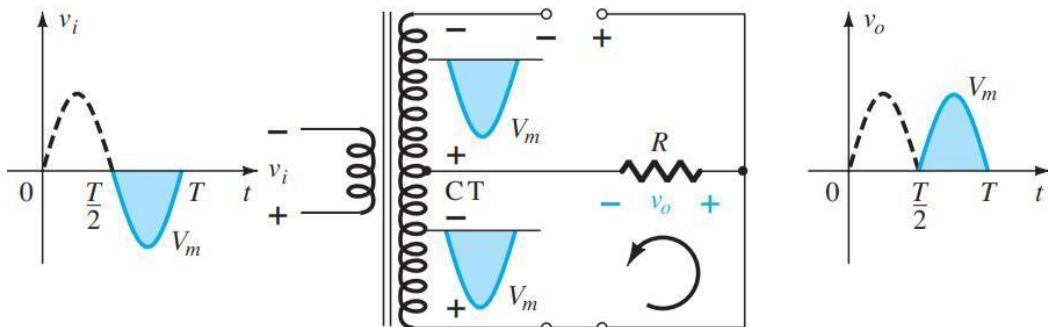


Figure 8: Network conditions for the negative region of v_i .

Apparatus

SL#	Apparatus	Quantity
1	Diode	1
2	Resistance ($1\text{k}\Omega$)	1
3	Capacitors ($47\mu\text{F}$ and $100\mu\text{F}$)	1 each
4	Transformer (220V/12V/9V/6V)	1
5	Project Board	1
6	Function Generator	1
7	Oscilloscope	1
8	Multimeter	1
9	Connecting Leads	10

Precaution:

1. A diode should never be removed or inserted into a circuit with voltage applied.
2. It should be ensured that a replacement diode is inserted into the circuit in the correct direction.
3. The correct connection of the transformer should be ensured.
4. When a diode is being tested, it should be ensured that the test voltage does not exceed the diode's:
 - a. Maximum allowable voltage.
 - b. It should be ensured that a replacement diode is inserted into the circuit in the correct direction.

Experimental Procedures:

1. The actual value of the $1\text{ k}\Omega$ resistor is measured.
2. The circuit is connected as shown in Figure 1 without any capacitors.
3. The AC power supply (function generator) is turned on with the voltage control knob set to 0 V.
4. The amplitude control knob is gradually rotated from 0 V to 10 V (maximum voltage).
5. The sinusoidal waveform is selected, and the frequency is set to 100 Hz.
6. The oscilloscope is connected to observe the wave shapes of the input and output voltages in dual channel mode with no capacitor connected across the load resistance, R.
7. The peak AC voltage is measured from the oscilloscope screen.
8. The output voltage is measured using a multimeter (in both DC and AC modes) and compared with the value obtained from the oscilloscope screen.
9. The measured data is recorded in Table 1.
10. The power supply is turned off, and $47\mu\text{F}$ and $100\mu\text{F}$ capacitors are connected across the load. The output voltage is observed and measured using both the oscilloscope

and multimeter, and the results are compared.

11. The power supply is turned off, and the measured data is recorded in Table 1.

The circuit is connected as shown in Figures 3 and 6 without any capacitors. Steps 3 to 11 are repeated, and the data is recorded in Tables 2 and 3, respectively.

Table 1: Data Table for the Circuit of Figure 1

Capacitance (μF)	Output Voltage, V_o (V) (By Oscilloscope)	Output Voltage, V_o (V) (By Multimeter)	
		DC Mode (V_{DC} , V)	AC Mode (V_{rms} , V)
0	9.2	2.84	3.54
47	8.95	8.88	0.051
100	8.9	8.89	0.041

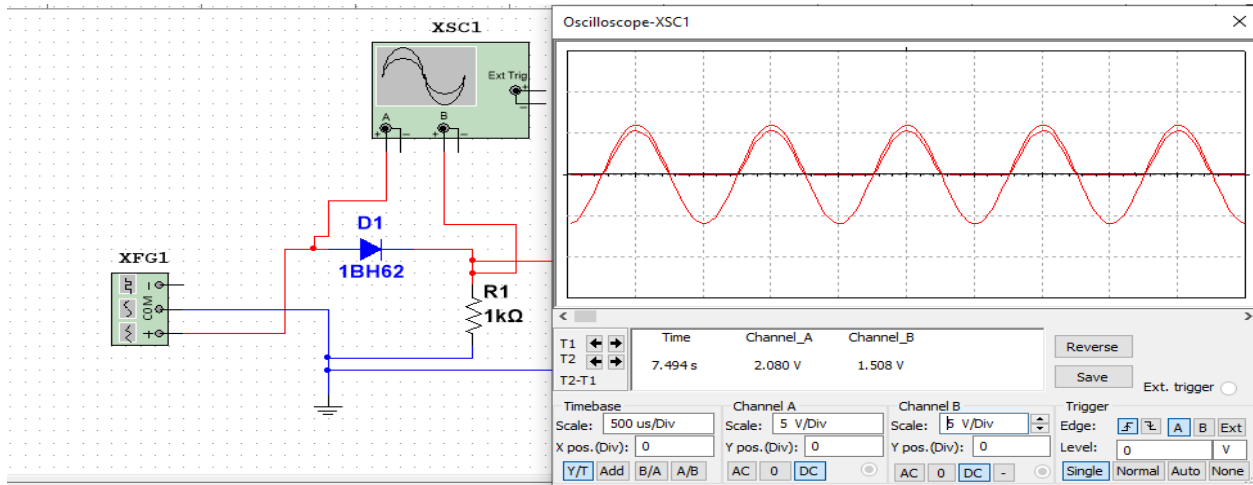
Table 2: Data Table for the Circuit of Figure 3

Capacitance (μF)	Output Voltage, V_o (V) (By Oscilloscope)	Output Voltage, V_o (V) (By Multimeter)	
		DC Mode (V_{DC} , V)	AC Mode (V_{rms} , V)
0	18.3	0.0015	14.14
47	19.9	0.0021	14.15
100	19.9	0.002	14.15

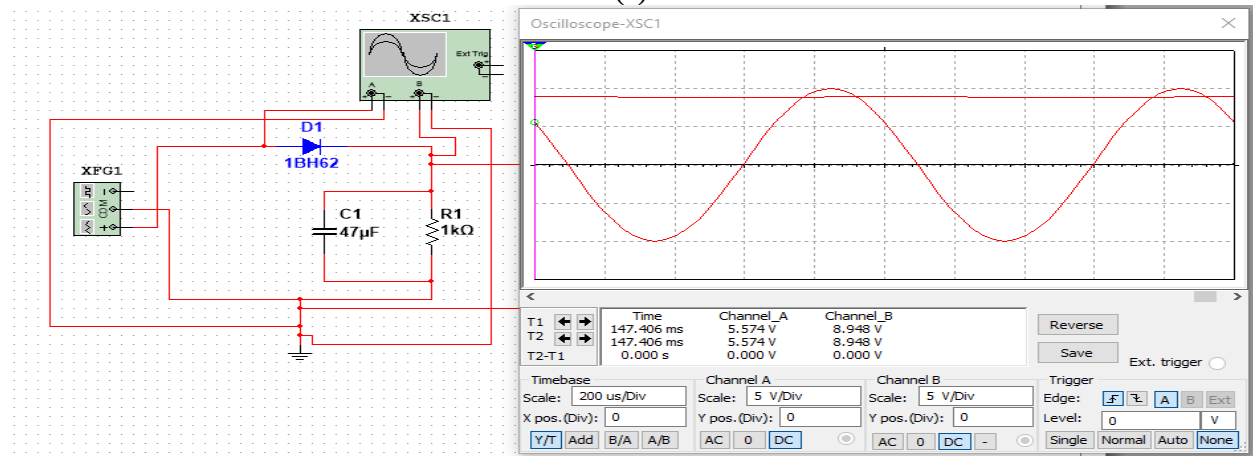
Table 3: Data Table for the Circuit of Figure 6

Capacitance (μF)	Output Voltage, V_o (V) (By Oscilloscope)	Output Voltage, V_o (V) (By Multimeter)	
		DC Mode (V_{DC} , V)	AC Mode (V_{rms} , V)
0	42.20	32.23	13.5
47	42.4	41.65	1.08
100	42.50	42.10	0.52

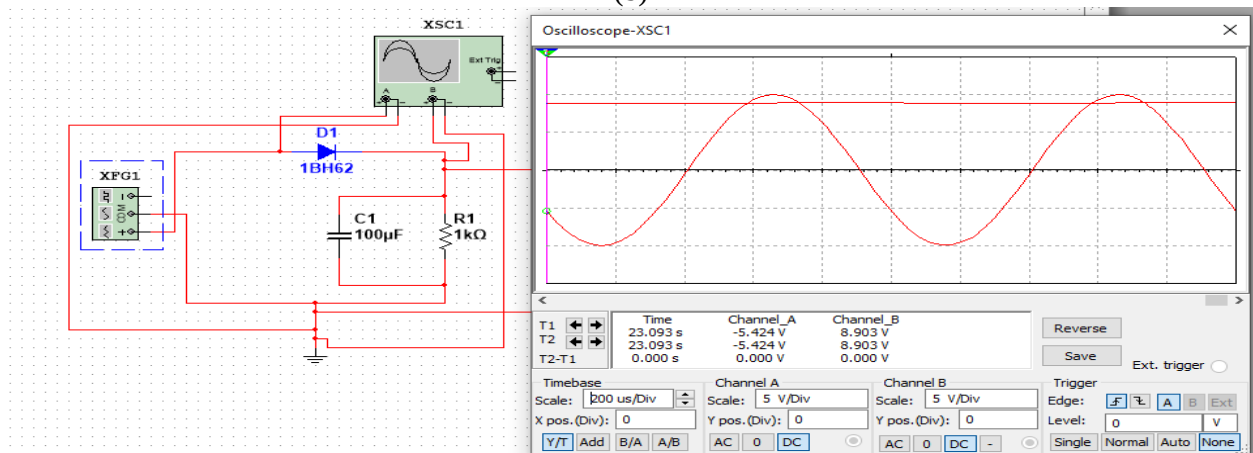
Simulation & Measurement



(a)

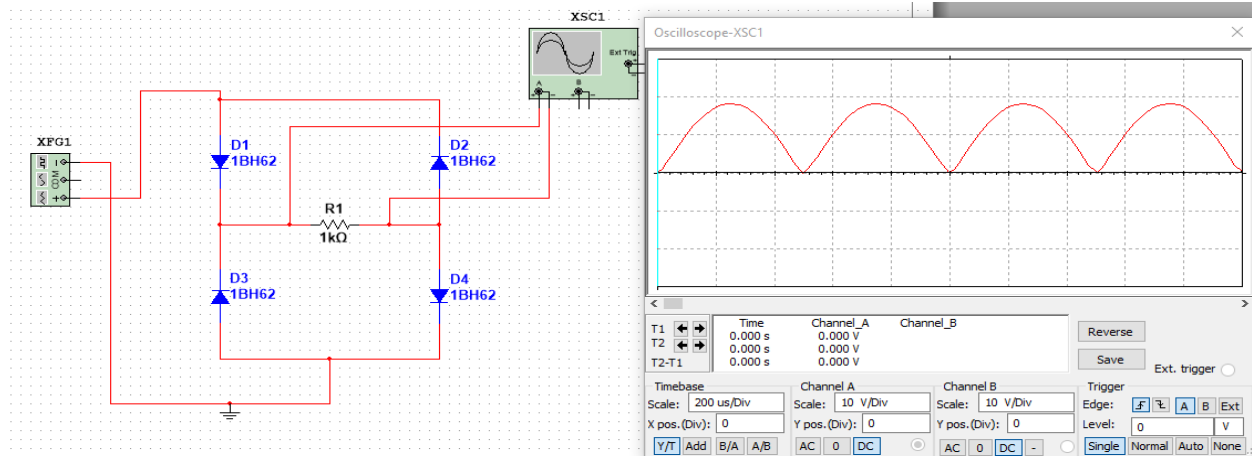


(b)

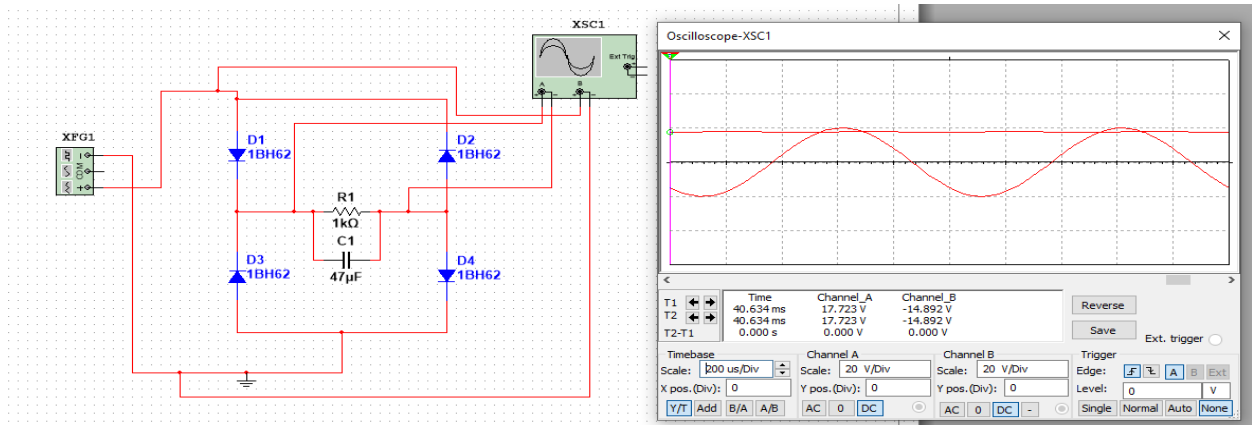


(c)

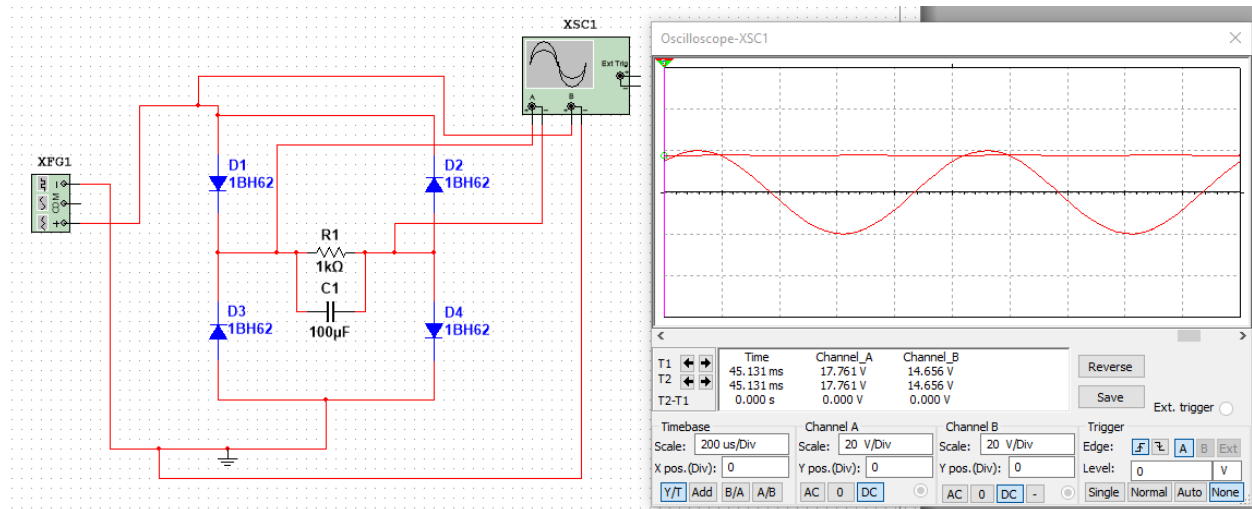
Fig 9: Simulation for the Circuit of Figure 1 (a)0μF (b)47μF (c)100μF



(a)

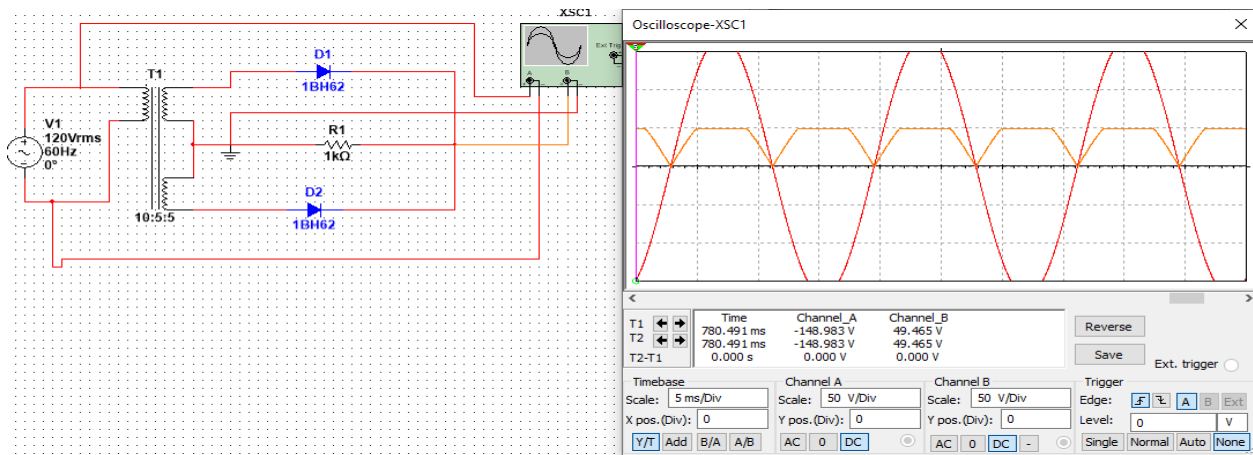


(b)

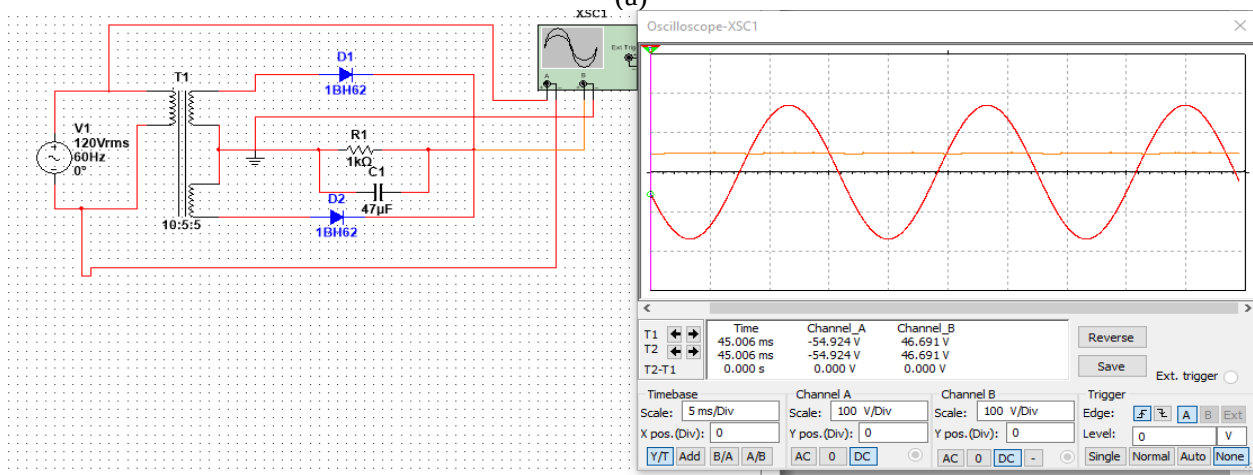


(c)

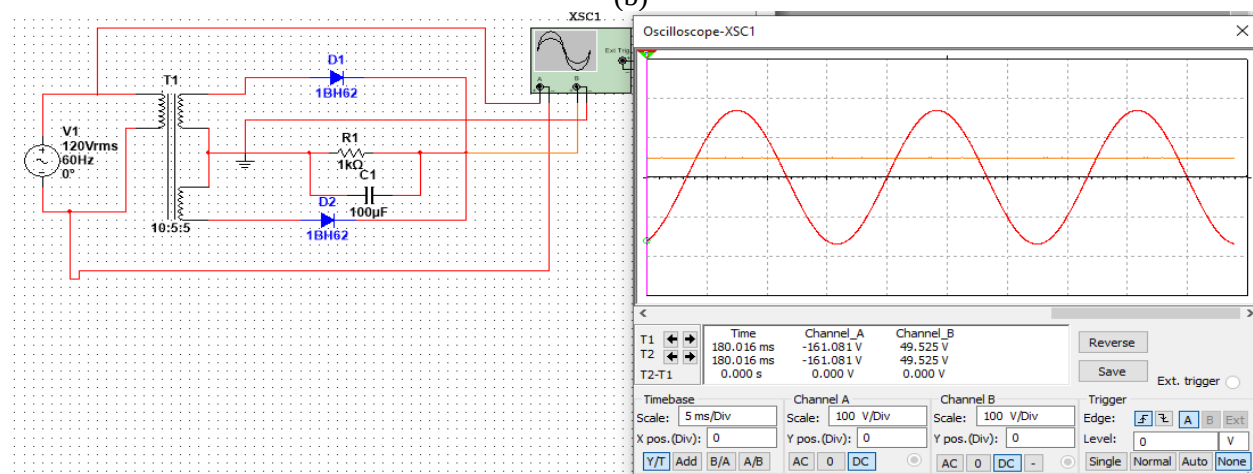
Fig 10: Simulation for the Circuit of Figure 3 (a)0μF (b)47μF (c)100μF



(a)



(b)



(c)

Fig 11: Simulation for the Circuit of Figure 3 (a)0μF (b)47μF (c)100μF

Answer of all questions

1. Show the difference between your simulated and measured values. Comment on the results.

Ans:

Observation: The measured V_{dc} for the Half-Wave Rectifier without a capacitor was 1.59V, while the peak voltage V_m was 5.3V.

Calculation: Theoretically, $V_{dc} = 0.318 \times V_m$. For, $V_m = 5.3$ V, the expected V_{dc} is approximately 1.68V.

Comment: There is a slight difference between the theoretical/simulated values and the measured values. This discrepancy is primarily due to the practical diode forward voltage drop (typically 0.7V for Silicon), internal resistance of the transformer, and the tolerance of the $1k\Omega$ resistor.

2. What are the effects and significance of using filter capacitance?

Ans:

Effects: Adding a capacitor (e.g., $47\mu F$ $100\mu F$) across the load resistance significantly increases the DC output voltage and reduces the AC ripple component.

Significance: The capacitor acts as a smoothing filter, storing charge during the conduction cycle and discharging it during the non-conduction period.

Results: As seen in Table 1, increasing the capacitance from $0\mu F$ to $100\mu F$ shifted the V_{dc} from 1.59V to 4.9V, effectively making the output nearly a constant DC voltage.

3. What is the minimum PIV for the diodes used in circuits 3 and 6?

Ans:

Circuit 3 (Bridge Rectifier): The minimum Peak Inverse Voltage (PIV) rating for each diode must be at least equal to the peak input voltage (V_m).

Circuit 6 (Center-Tapped Rectifier): In this configuration, each diode is subjected to twice the peak secondary voltage of one-half of the transformer. Therefore, the minimum PIV must be $2V_m$.

4. Why is the circuit in Fig. 6 better than that in Fig. 3?

Ans:

- **Voltage Drop:** In the Bridge Rectifier (Fig. 3), current flows through two diodes simultaneously in each half-cycle, resulting in a total diode pressure drop of approximately 1.4V.
- **Efficiency:** The Center-Tapped Rectifier (Fig. 6) only has one diode conducting at a time, which reduces the internal voltage drop to 0.7V, making it more efficient for low-voltage applications

5. Discuss the overall aspects of the experiment. Did your results match the expected ones?

Ans:

This experiment demonstrated how semiconductor diodes can rectify AC into DC. We observed that full-wave rectification provides double the average DC level compared to half-wave rectification (0.636Vm vs 0.318Vm).

Discussion

In this experiment, we observed how different rectifier circuits convert AC to DC and how capacitors help smooth the output. For the half-wave rectifier (Figure 1), the AC ripple was high without a capacitor, but adding capacitors (47 μ F and 100 μ F) greatly reduced it. In the full-wave bridge rectifier (Figure 3), adding capacitors lowered the AC ripple even further, providing a more stable DC output. The center-tapped full-wave rectifier (Figure 6) also showed a smoother output with capacitors.

Overall, adding capacitance reduced ripple and made the DC output steadier, especially in the full-wave rectifiers.

Conclusion

This experiment successfully demonstrated how different rectifier circuits—half-wave, full-wave bridge, and center-tapped full-wave—convert AC to DC and the importance of filtering capacitors in achieving a stable DC output. We observed that full-wave rectifiers provide a smoother output than half-wave, especially when capacitors are added to reduce AC ripple. Increasing capacitance further stabilized the DC output, highlighting the capacitor's role in filtering. Overall, the experiment helped us understand rectifier operation and how filtering improves DC voltage quality in electronic circuits.

References

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